

TRIGONOMETRIC RATIO

JEE EXERCISES

Exercise J-1

INTRODUCTION TO TRIGONOMETRY RATIOS

1. The expression $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$ can be written as : [JEE(Main)-2013]

व्यंजक $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$ को लिखा जा सकता है : [JEE(Main)-2013]

- (A) $\sin A \cos A + 1$ (B) $\sec A \operatorname{cosec} A + 1$ (C) $\tan A + \cot A$ (D) $\sec A + \operatorname{cosec} A$

Ans. B

Sol. Given expression

$$\begin{aligned} &= \frac{\sin A}{\cos A} \times \frac{\sin A}{\sin A - \cos A} + \frac{\cos A}{\sin A} \times \frac{\cos A}{\cos A - \sin A} \\ &= \frac{1}{\sin A - \cos A} \left\{ \frac{\sin^3 A - \cos^3 A}{\cos A \sin A} \right\} \\ &= \frac{\sin^2 A + \sin A \cos A + \cos^2 A}{\sin A \cos A} = 1 + \sec A \operatorname{cosec} A \end{aligned}$$

2. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ where $x \in \mathbb{R}$ and $k \geq 1$. Then $f_4(x) - f_6(x)$ equals : [JEE(Main)-2014]

माना $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ जहाँ $x \in \mathbb{R}$ तथा $k \geq 1$ तो $f_4(x) - f_6(x)$ का मान होगा : [JEE(Main)-2014]

- (A) $1/4$ (B) $1/12$ (C) $1/6$ (D) $1/3$

Ans. B

Sol. $f_k(x) = \frac{1}{k} \{ \sin^k x + \cos^k x \}$

$$\begin{aligned} \therefore f_4(x) - f_6(x) &= \frac{1}{4} \{ \sin^4 x + \cos^4 x \} - \frac{1}{6} \{ \sin^6 x + \cos^6 x \} \\ &= \frac{1}{4} \{ 1 - 2 \sin^2 x \cos^2 x \} - \frac{1}{6} \{ 1 - 3 \sin^2 x \cos^2 x \} = \frac{1}{12} \end{aligned}$$

BASED ON COMPOUND ANGLES

3. If $a \cos \theta + b \sin \theta = 3$ and $a \sin \theta - b \cos \theta = 4$, then the value of $a^2 + b^2$ is :

- (A) 9 (B) 16 (C) 25 (D) None of these

यदि $a \cos \theta + b \sin \theta = 3$ एवं $a \sin \theta - b \cos \theta = 4$ तब $a^2 + b^2$ का मान होगा :

- (A) 9 (B) 16 (C) 25 (D) इनमें से कोई नहीं

Ans. C

Sol. $a \cos \theta + b \sin \theta = 3$
 $\Rightarrow (a \cos \theta + b \sin \theta)^2 = 9$
 $\Rightarrow a^2 \cos^2 \theta + b^2 \sin^2 \theta + 2ab \sin \theta \cos \theta = 9 \quad \dots(i)$
 $a \sin \theta - b \cos \theta = 4$
 $\Rightarrow (a \sin \theta - b \cos \theta)^2 = 16$
 $\Rightarrow a^2 \sin^2 \theta + b^2 \cos^2 \theta - 2ab \sin \theta \cos \theta = 16 \quad \dots(ii)$
 By (i) + (ii); we get
 $a^2 + b^2 = 25$

4. The expression $\sec^2 \theta \cdot \operatorname{cosec}^2 \theta$ is equal to :

- (A) $\sin^2 \theta + \cos^2 \theta$ (B) $\sin^2 \theta - \cos^2 \theta$ (C) $\sec^2 \theta + \operatorname{cosec}^2 \theta$ (D) None of these

यंजक $\sec^2 \theta \cdot \operatorname{cosec}^2 \theta$ किस के तुल्य है :

- (A) $\sin^2 \theta + \cos^2 \theta$ (B) $\sin^2 \theta - \cos^2 \theta$ (C) $\sec^2 \theta + \operatorname{cosec}^2 \theta$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\sec^2 \theta \operatorname{cosec}^2 \theta = \frac{1}{\sin^2 \theta \cos^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta}$
 $= \sec^2 \theta + \operatorname{cosec}^2 \theta$

5. If $\tan 50^\circ - \tan 40^\circ = k \tan 10^\circ$, then the value of k is :

- (A) 2 (B) 1 (C) 3 (D) None of these

यदि $\tan 50^\circ - \tan 40^\circ = k \tan 10^\circ$ तब k का मान क्या होगा :

- (A) 2 (B) 1 (C) 3 (D) इनमें से कोई नहीं

Ans. A

Sol. $\because \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
 $\Rightarrow \tan 50^\circ - \tan 40^\circ = \tan(50^\circ - 40^\circ)(1 + \tan 50^\circ \tan 40^\circ)$
 $\therefore k = 1 + \tan 50^\circ \tan 40^\circ = \frac{\cos 40^\circ \cos 50^\circ + \sin 40^\circ \sin 50^\circ}{\cos 40^\circ \cos 50^\circ}$

$$= \frac{2 \cos(50^\circ - 40^\circ)}{2 \cos 40^\circ \cos 50^\circ} = \frac{2 \cos(10^\circ)}{\cos(40^\circ + 50^\circ) + \cos(40^\circ - 50^\circ)}$$

$$\Rightarrow K = \frac{2 \cos 10^\circ}{0 + \cos 10^\circ} = 2$$

6. The exact value of $\frac{\sin 24^\circ \cos 6^\circ - \sin 6^\circ \sin 66^\circ}{\sin 21^\circ \cos 39^\circ - \cos 51^\circ \sin 69^\circ}$ is :

- (A) 0 (B) 1 (C) -1 (D) None of these

$$\frac{\sin 24^\circ \cos 6^\circ - \sin 6^\circ \sin 66^\circ}{\sin 21^\circ \cos 39^\circ - \cos 51^\circ \sin 69^\circ} \text{ का निश्चित मान क्या होगा :}$$

- (A) 0 (B) 1 (C) -1 (D) इनमें से कोई नहीं

Ans. C

Sol.

$$\frac{\sin 24^\circ \cos 6^\circ - \sin 6^\circ \sin 66^\circ}{\sin 21^\circ \cos 39^\circ - \cos 51^\circ \sin 69^\circ}$$

$$= \frac{\sin 24^\circ \cos 6^\circ - \sin 6^\circ \sin(90^\circ - 24^\circ)}{\sin 21^\circ \cos 39^\circ - \cos(90^\circ - 39^\circ) \sin(90^\circ - 21^\circ)}$$

$$= \frac{\sin 24^\circ \cos 6^\circ - \sin 6^\circ \cos 24^\circ}{\sin 21^\circ \cos 39^\circ - \cos(21^\circ) \sin(39^\circ)}$$

$$= \frac{\sin(24^\circ - 6^\circ)}{\sin(21^\circ - 39^\circ)} = \frac{\sin(18^\circ)}{\sin(-18^\circ)} = -1$$

7. If $\tan \alpha + \cot \alpha = a$, then the value of $\tan^4 \alpha + \cot^4 \alpha$ is :

- (A) $a^2 - 4a + 2$ (B) $a^4 - 4a^2 + 2$ (C) $a^4 + 4a^2 - 2$ (D) None of these

यदि $\tan \alpha + \cot \alpha = a$ तब $\tan^4 \alpha + \cot^4 \alpha$ किस के तुल्य होगा :

- (A) $a^2 - 4a + 2$ (B) $a^4 - 4a^2 + 2$ (C) $a^4 + 4a^2 - 2$ (D) इनमें से कोई नहीं

Ans. B

Sol. Given $\tan \alpha + \cot \alpha = a$

$$\tan^4 \alpha + \cot^4 \alpha = (\tan^2 \alpha + \cot^2 \alpha)^2 - 2 \tan^2 \alpha \cot^2 \alpha = [(\tan \alpha + \cot \alpha)^2 - 2]^2 - 2$$

$$= (a^2 - 2)^2 - 2$$

$$= a^4 - 4a^2 + 4 - 2$$

$$= a^4 - 4a^2 + 2$$

8. The simplified value of, $\frac{2(\sin 2\theta + 2\cos^2 \theta - 1)}{\cos \theta - \sin \theta - \cos 3\theta + \sin 3\theta}$ is :

$\frac{2(\sin 2\theta + 2\cos^2 \theta - 1)}{\cos \theta - \sin \theta - \cos 3\theta + \sin 3\theta}$ का सरलतम रूप क्या होगा :

- (A) $\sin \theta$ (B) $\cos \theta$ (C) $\operatorname{cosec} \theta$ (D) $\sec \theta$

Ans. C

Sol.

$$\begin{aligned} & \frac{2(\sin 2\theta + 2\cos^2 \theta - 1)}{\cos \theta - \sin \theta - \cos 3\theta + \sin 3\theta} \\ &= \frac{2(\sin 2\theta + \cos 2\theta)}{\cos \theta - \cos 3\theta + \sin 3\theta - \sin \theta} \\ &= \frac{2(\sin 2\theta + \cos 2\theta)}{2\sin 2\theta \sin \theta + 2\cos 2\theta \sin \theta} \\ &= \frac{2(\sin 2\theta + \cos 2\theta)}{2\sin \theta (\sin 2\theta + \cos 2\theta)} = \operatorname{cosec} \theta \end{aligned}$$

9. If $\cos \theta = \frac{1}{2}\left(a + \frac{1}{a}\right)$, then the value of $\cos 3\theta$ in terms of 'a' is :

- (A) $\frac{1}{4}\left(a^3 + \frac{1}{a^3}\right)$ (B) $\frac{1}{2}\left(a^3 + \frac{1}{a^3}\right)$ (C) $4\left(a^3 + \frac{1}{a^3}\right)$ (D) None of these

यदि $\cos \theta = \frac{1}{2}\left(a + \frac{1}{a}\right)$ तब $\cos 3\theta$ का मान 'a' के रूप में क्या होगा :

- (A) $\frac{1}{4}\left(a^3 + \frac{1}{a^3}\right)$ (B) $\frac{1}{2}\left(a^3 + \frac{1}{a^3}\right)$ (C) $4\left(a^3 + \frac{1}{a^3}\right)$ (D) इनमें से कोई नहीं

Ans. B

Sol.

$$\begin{aligned} \cos \theta &= \frac{1}{2}\left(a + \frac{1}{a}\right) \\ \Rightarrow \cos 3\theta &= 4\cos^3 \theta - 3\cos \theta \\ &= \cos \theta (4\cos^2 \theta - 3) \end{aligned}$$

$$\begin{aligned}
&= \left(a + \frac{1}{a} \right) \frac{1}{2} \left[4 \left(\frac{1}{2} \left(a + \frac{1}{a} \right) \right)^2 - 3 \right] \\
&= \frac{1}{2} \left(a + \frac{1}{a} \right) \left[a^2 + \frac{1}{a^2} + 2 - 3 \right] \\
&= \frac{1}{2} \left(a + \frac{1}{a} \right) \left(a^2 + \frac{1}{a^2} - 1 \right) \\
&= \frac{1}{2} \left(a^3 + \frac{1}{a} - a + a + \frac{1}{a^3} - \frac{1}{a} \right) = \frac{1}{2} \left(a^3 + \frac{1}{a^3} \right)
\end{aligned}$$

10. $(1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2$ is equal to :

$(1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2$ किस के बराबर है :

(A) $\tan^2 \alpha + \tan^2 \beta$ (B) $\cos^2 \alpha \cos^2 \beta$ (C) $\sec^2 \alpha \sec^2 \beta$ (D) $\tan^2 \alpha \tan^2 \beta$

Ans. C

Sol. $(1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2$

$$= 1 + \tan^2 \alpha \tan^2 \beta + 2 \tan \alpha \tan \beta + \tan^2 \alpha + \tan^2 \beta - 2 \tan \alpha \tan \beta$$

$$= 1 + \tan^2 \alpha + \tan^2 \beta (1 + \tan^2 \alpha)$$

$$= (1 + \tan^2 \alpha) (1 + \tan^2 \beta)$$

$$= \sec^2 \alpha \sec^2 \beta$$

11. If $\frac{1 + \sin 2x}{1 - \sin 2x} = \cot^2 (a + x) \forall x \in \mathbb{R} - \left(n\pi + \frac{\pi}{4} \right), n \in \mathbb{N}$. Then 'a' equals :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) None of these

यदि $\frac{1 + \sin 2x}{1 - \sin 2x} = \cot^2 (a + x) \forall x \in \mathbb{R} - \left(n\pi + \frac{\pi}{4} \right), n \in \mathbb{N}$ तब 'a' किस के बराबर होगा :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) इनमें से कोई नहीं

Ans. C

Sol. Given

$$\frac{1 + \sin 2x}{1 - \sin 2x} = \cot^2(a + x) \quad \forall x \in \mathbb{R} - \left\{ n\pi + \frac{\pi}{4} \right\}$$

$$\Rightarrow \frac{\sin^2 x + \cos^2 x + 2 \sin x \cos x}{\sin^2 x + \cos^2 x - 2 \sin x \cos x} = \cot^2(a + x)$$

$$\Rightarrow \left[\frac{\sin x + \cos x}{\sin x - \cos x} \right]^2 = \cot^2(a + x)$$

$$\Rightarrow \left(\frac{1 + \tan x}{\tan x - 1} \right)^2 = \frac{1}{\tan^2(a + x)}$$

$$\Rightarrow \tan^2\left(\frac{\pi}{4} + x\right) \cdot \tan^2(a + x) = 1$$

$$\Rightarrow \left[\tan\left(\frac{\pi}{4} + x\right) \cdot \tan(a + x) \right]^2 = 1$$

$$\Rightarrow a = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

12. If $\tan \alpha = \frac{5}{6}$ and $\tan \beta = \frac{1}{11}$, then $\left(0 < \alpha, \beta < \frac{\pi}{2} \right)$:

(A) $\alpha + \beta = \frac{\pi}{6}$ (B) $\alpha + \beta = \frac{\pi}{4}$ (C) $\alpha + \beta = \frac{\pi}{3}$ (D) None of these

यदि $\tan \alpha = \frac{5}{6}$ एवं $\tan \beta = \frac{1}{11}$, तब $\left(0 < \alpha, \beta < \frac{\pi}{2} \right)$:

(A) $\alpha + \beta = \frac{\pi}{6}$ (B) $\alpha + \beta = \frac{\pi}{4}$ (C) $\alpha + \beta = \frac{\pi}{3}$ (D) इनमें से कोई नहीं

Ans. B

Sol. Given $\tan \alpha = \frac{5}{6}$ $\tan \beta = \frac{1}{11}$ $\left(0 < \alpha, \beta < \frac{\pi}{2} \right)$

$$\therefore \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{\frac{5}{6} + \frac{1}{11}}{1 - \frac{5}{66}} = \frac{61}{61} = 1$$

$$\Rightarrow \alpha + \beta = \frac{\pi}{4} \quad \{\because 0 < \alpha + \beta < \pi\}$$

13. If $\tan x \tan y = a$ and $x + y = \frac{\pi}{6}$, then $\tan x$ and $\tan y$ satisfy the equation :

यदि $\tan x \tan y = a$ एवं $x + y = \frac{\pi}{6}$ तब $\tan x$ एवं $\tan y$ निम्न में से किस समीकरण को संतुष्ट करेंगे :

(A) $x^2 - \sqrt{3}(1-a)x + a = 0$

(B) $\sqrt{3}x^2 - (1-a)x + a\sqrt{3} = 0$

(C) $x^2 + \sqrt{3}(1-a)x - a = 0$

(D) $\sqrt{3}x^2 + (1+a)x - a\sqrt{3} = 0$

Ans. B

Sol. Given

$$x + y = \frac{\pi}{6} \text{ \& \; } \tan x \tan y = a$$

Let equation whose roots will be $\tan x$ & $\tan y$ is

$$t^2 - (\tan x + \tan y)t + \tan x \tan y = 0$$

$$t^2 - (\tan(x+y)(1 - \tan x \tan y))t + a = 0$$

$$\Rightarrow t^2 - \left(\tan\left(\frac{\pi}{6}\right)(1-a) \right)t + a = 0$$

$$\Rightarrow t^2 - \left(\frac{1-a}{\sqrt{3}} \right)t + a = 0$$

$$\Rightarrow \sqrt{3}t^2 - (1-a)t + a\sqrt{3} = 0$$

$$\Rightarrow \text{equation is } \sqrt{3}x^2 - (1-a)x + \sqrt{3}a = 0$$

14. Which of the following number(s) is/are rational?

निम्न में से कौनसी संख्या परिमेय है?

(A) $\sin 15^\circ$

(B) $\cos 15^\circ$

(C) $\sin 15^\circ \cos 15^\circ$

(D) $\sin 15^\circ \cos 75^\circ$

Ans. C

Sol. (A) $\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}} \in \mathbb{Q}^c$ (Irrational)

(B) $\cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}} \in \mathbb{Q}^c$ (Irrational)

(C) $\sin 15^\circ \cos 15^\circ = \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right) = \frac{1}{4} \in \mathbb{Q}$ (Rational)

(D) $\sin 15^\circ \cos 75^\circ = \sin^2 15^\circ = \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2 \in \mathbb{Q}^c$ (Irrational)

15. If $\cos x + \cos y + \cos \alpha = 0$ and $\sin x + \sin y + \sin \alpha = 0$, then $\cot\left(\frac{x+y}{2}\right) =$

यदि $\cos x + \cos y + \cos \alpha = 0$ तथा $\sin x + \sin y + \sin \alpha = 0$, तब $\cot\left(\frac{x+y}{2}\right) =$

(A) $\sin \alpha$ (B) $\cos \alpha$ (C) $\cot \alpha$ (D) $2 \sin \alpha$

Ans. C

Sol. Given $\cos x + \cos y + \cos \alpha = 0$
& $\sin x + \sin y + \sin \alpha = 0$

$$\Rightarrow 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) = -\cos \alpha$$

$$\& 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) = -\sin \alpha$$

$$\cot\left(\frac{x+y}{2}\right) = \cot \alpha$$

16. If the roots of the quadratic equation, $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$ respectively, then the value of $2 + q - p$ is : **[AIEEE-2006]**

द्विघात समीकरण $x^2 + px + q = 0$ के मूल $\tan 30^\circ$ एवं $\tan 15^\circ$ हैं। तब $2 + q - p$ का मान क्या होगा : **[AIEEE-2006]**

(A) 2 (B) 3 (C) 0 (D) 1

Ans. B

Sol. $x^2 + px + q = 0$
 $\tan 30^\circ + \tan 15^\circ = -p$
 $\tan 30^\circ \cdot \tan 15^\circ = q$

$$\tan 45^\circ = \frac{\tan 30^\circ + \tan 15^\circ}{1 - \tan 30^\circ \tan 15^\circ} = \frac{-p}{1-q} = 1$$

$$\Rightarrow -p = 1 - q$$

$$\Rightarrow q - p = 1 \quad \therefore 2 + q - p = 3$$

17. Let $\cos(\alpha + \beta) = \frac{4}{5}$ and let $\sin(\alpha - \beta) = \frac{5}{13}$, where $0 \leq \alpha, \beta \leq \frac{\pi}{4}$. Then $\tan 2\alpha =$ [AIEEE-2010]

माना $\cos(\alpha + \beta) = \frac{4}{5}$ तथा माना $\sin(\alpha - \beta) = \frac{5}{13}$, जहाँ $0 \leq \alpha, \beta \leq \frac{\pi}{4}$ हैं तो $\tan 2\alpha =$ [AIEEE-2010]

(A) $\frac{25}{16}$ (B) $\frac{56}{33}$ (C) $\frac{19}{12}$ (D) $\frac{20}{7}$

Ans. B

Sol. Given $\cos(\alpha + \beta) = \frac{4}{5}$ $\sin(\alpha - \beta) = \frac{5}{13}$ $0 \leq \alpha, \beta \leq \frac{\pi}{4}$

$$\Rightarrow \tan(\alpha + \beta) = \frac{3}{4} \text{ \& } \tan(\alpha - \beta) = \frac{5}{12}$$

$$\Rightarrow \tan(2\alpha) = \tan(\alpha + \beta + \alpha - \beta)$$

$$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta)\tan(\alpha - \beta)}$$

$$= \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{5}{16}} = \frac{14}{12} \times \frac{16}{11}$$

$$= \frac{56}{33}$$

18. In a ΔPQR , if $3 \sin P + 4 \cos Q = 6$ and $4 \sin Q + 3 \cos P = 1$, then the angle R is equal to : [AIEEE-2012]

ΔPQR में यदि $3 \sin P + 4 \cos Q = 6$ तथा $4 \sin Q + 3 \cos P = 1$ है, तो कोण R बराबर हैं : [AIEEE-2012]

(A) $\frac{3\pi}{4}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

Ans. C

Sol. $3\sin P + 4\cos Q = 6$ (1)

$$4\sin Q + 3\cos P = 1 \dots (2)$$

Squaring (1) and (2). Then adding

$$9 + 16 + 24\sin(P + Q) = 37$$

$$\Rightarrow 24\sin(P + Q) = 12$$

$$\Rightarrow \sin(P + Q) = \frac{1}{2} \Rightarrow \sin R = \frac{1}{2}$$

$$\therefore R = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

But $R = \frac{5\pi}{6}$ doesn't satisfy the given equation so

$$R = \frac{\pi}{6}$$

COMPONENDO AND DIVIDENDO

19. If $3 \sin \beta = \sin(2\alpha + \beta)$ then :

(A) $\tan(\alpha + \beta) = 2 \tan \alpha$

(B) $2 \sin \beta = \sin(\alpha - \beta) \cos \alpha$

(C) $\sin \beta = \cos(\alpha + \beta) \cos \alpha$

(D) None of these

यदि $3 \sin \beta = \sin(2\alpha + \beta)$ तब :

(A) $\tan(\alpha + \beta) = 2 \tan \alpha$

(B) $2 \sin \beta = \sin(\alpha - \beta) \cos \alpha$

(C) $\sin \beta = \cos(\alpha + \beta) \cos \alpha$

(D) इनमें से कोई नहीं

Ans. A

Sol. $3 \sin \beta = \sin(2\alpha + \beta)$

$$\Rightarrow \frac{\sin(2\alpha + \beta)}{\sin \beta} = \frac{3}{1}$$

Apply componendo and dividendo

$$\frac{\sin(2\alpha + \beta) + \sin \beta}{\sin(\beta + 2\alpha) - \sin \beta} = \frac{4}{2}$$

$$\frac{2 \sin(\alpha + \beta) \cos \alpha}{2 \cos(\alpha + \beta) \sin(\alpha)} = 2$$

$$\Rightarrow \tan(\alpha + \beta) = 2 \tan \alpha$$

20. If $\sin(x + 3\alpha) = 3 \sin(\alpha - x)$, then :

यदि $\sin(x + 3\alpha) = 3 \sin(\alpha - x)$ तब :

- (A) $\tan x = \tan \alpha$ (B) $\tan x = \tan^2 \alpha$ (C) $\tan x = \tan^3 \alpha$ (D) $\tan x = 3 \tan \alpha$

Ans. C

Sol. $\sin(x + 3\alpha) = 3 \sin(\alpha - x)$

$$\Rightarrow \frac{\sin(x + 3\alpha)}{\sin(\alpha - x)} = 3$$

Apply componendo and dividendo

$$\Rightarrow \frac{\sin(x + 3\alpha) + \sin(\alpha - x)}{\sin(x + 3\alpha) - \sin(\alpha - x)} = \frac{4}{2} = 2$$

$$\Rightarrow \frac{2 \sin(2\alpha) \cos(x + \alpha)}{2 \cos(2\alpha) \sin(x + \alpha)} = 2$$

$$\Rightarrow 2 \tan(x + \alpha) = \tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\Rightarrow \left(\frac{\tan x + \tan \alpha}{1 - \tan x \tan \alpha} \right) \cdot 2 = 2 \left(\frac{\tan \alpha}{1 - \tan^2 \alpha} \right)$$

$$\tan \alpha - \tan x \tan^2 \alpha = \tan x + \tan \alpha - \tan^2 \alpha \tan x - \tan^3 \alpha$$

$$\Rightarrow \tan x = \tan^3 \alpha$$

21. If $\cos x = \frac{2 \cos y - 1}{2 - \cos y}$, where $x, y \in (0, \pi)$. Then $\tan \frac{x}{2} \cdot \cot \frac{y}{2}$ is equal to :

यदि $\cos x = \frac{2 \cos y - 1}{2 - \cos y}$, जहाँ $x, y \in (0, \pi)$. तब $\tan \frac{x}{2} \cdot \cot \frac{y}{2}$ बराबर होगा :

- (A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{\sqrt{3}}$

Ans. B

Sol. $\frac{1}{\cos x} = \frac{2 - \cos y}{2 \cos y - 1}$ Applying C/D $\Rightarrow \frac{1 - \cos x}{1 + \cos x} = \frac{3(1 - \cos y)}{1 + \cos y}$

$$\Rightarrow \tan^2 \frac{x}{2} = 3 \tan^2 \frac{y}{2} \Rightarrow \tan^2 \frac{x}{2} \cot^2 \frac{y}{2} = 3$$

CONTINUED PRODUCT

22. $\left(1 + \cos \frac{\pi}{8}\right)\left(1 + \cos \frac{3\pi}{8}\right)\left(1 + \cos \frac{5\pi}{8}\right)\left(1 + \cos \frac{7\pi}{8}\right)$ is equal to :

$\left(1 + \cos \frac{\pi}{8}\right)\left(1 + \cos \frac{3\pi}{8}\right)\left(1 + \cos \frac{5\pi}{8}\right)\left(1 + \cos \frac{7\pi}{8}\right)$ बराबर होगा :

- (A) $\frac{1}{2}$ (B) $\cos \frac{\pi}{8}$ (C) $\frac{1}{8}$ (D) $\frac{1 + \sqrt{2}}{2\sqrt{2}}$

Ans. C

Sol.

$$\begin{aligned} & \left(1 + \cos \frac{\pi}{8}\right)\left(1 + \cos \frac{3\pi}{8}\right)\left(1 + \cos \frac{5\pi}{8}\right)\left(1 + \cos \frac{7\pi}{8}\right) \\ &= \left(1 + \cos \frac{\pi}{8}\right)\left(1 + \cos \frac{3\pi}{8}\right)\left(1 + \cos \left(\pi - \frac{3\pi}{8}\right)\right)\left(1 + \cos \left(\pi - \frac{\pi}{8}\right)\right) \\ &= \left(1 + \cos \frac{\pi}{8}\right)\left(1 - \cos \frac{\pi}{8}\right)\left(1 + \cos \frac{3\pi}{8}\right)\left(1 - \cos \frac{3\pi}{8}\right) \\ &= \left(1 - \cos^2 \frac{\pi}{8}\right)\left(1 - \cos^2 \frac{3\pi}{8}\right) = \left[\sin \frac{\pi}{8} \sin \left(\frac{3\pi}{8}\right)\right]^2 \\ &= \left[\sin \left(\frac{\pi}{8}\right) \sin \left(\frac{\pi}{2} - \frac{\pi}{8}\right)\right]^2 = \left[\sin \frac{\pi}{8} \cos \frac{\pi}{8}\right]^2 \\ &= \frac{1}{4} \left[\sin \left(2 \cdot \frac{\pi}{8}\right)\right]^2 = \frac{1}{4} \left[\sin \frac{\pi}{4}\right]^2 = \frac{1}{8} \end{aligned}$$

23. $\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 179^\circ =$

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol.

$$\begin{aligned} & \cos 1^\circ \cos 2^\circ \dots \cos (179^\circ) \\ &= \cos 1^\circ \cos 2^\circ \dots \cos 89^\circ \cos 90^\circ \dots \cos (179^\circ) \\ &= 0 \end{aligned}$$

RANGE OF TRIGONOMETRIC FUNCTIONS

24. If $A = \sin^2 x + \cos^4 x$, then for all real x : [AIEEE-2011]

यदि $A = \sin^2 x + \cos^4 x$, तो सभी वास्तविक x के लिये मान होंगे : [AIEEE-2011]

- (A) $1 \leq A \leq 2$ (B) $\frac{3}{4} \leq A \leq \frac{13}{16}$ (C) $\frac{3}{4} \leq A \leq 1$ (D) $\frac{13}{16} \leq A \leq 1$

Ans. C

Sol. $A = \sin^2 x + \cos^4 x$

$$= \cos^4 x - \cos^2 x + 1$$

$$= (\cos^2 x)^2 - 2 \cdot \cos^2 x \cdot \frac{1}{2} + \frac{1}{4} - \frac{1}{4} + 1$$

$$= (\cos^2 x - \frac{1}{2})^2 + \frac{3}{4}$$

$$\because 0 \leq \cos^2 x \leq 1$$

$$\Rightarrow -\frac{1}{2} \leq \cos^2 x - \frac{1}{2} \leq \frac{1}{2}$$

$$\Rightarrow 0 \leq \left(\cos^2 x - \frac{1}{2} \right)^2 \leq \frac{1}{4}$$

$$\Rightarrow \frac{3}{4} \leq A \leq 1$$

SUMMATION OF SERIES

25. The sum of $\sin^3 \theta + \frac{1}{3} \sin^3 3\theta + \frac{1}{3^2} \sin^3 9\theta + \dots +$ to n terms is :

$\sin^3 \theta + \frac{1}{3} \sin^3 3\theta + \frac{1}{3^2} \sin^3 9\theta + \dots + n$ के पदों का मान होगा :

(A) $\frac{1}{2} \left[3 \sin \theta - \frac{1}{3^{n-1}} \sin (3^n \theta) \right]$

(B) $\frac{1}{2} \left[3 \cos \theta - \frac{1}{3^{n-1}} \sin (3^n \theta) \right]$

(C) $\frac{1}{4} \left[3 \cos \theta - \frac{1}{3^{n-1}} \sin (3^n \theta) \right]$

(D) $\frac{1}{4} \left[3 \sin \theta - \frac{1}{3^{n-1}} \sin (3^n \theta) \right]$

Ans. D

Sol. $\because \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$

$$\Rightarrow \sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4}$$

$$\Rightarrow \sin^3 \theta + \frac{1}{3} \sin^3 (3\theta) + \frac{1}{3^2} \sin^3 (9\theta) + \dots + \frac{1}{3^{n-1}} \sin^3 (3^{n-1} \theta)$$

$$= \left(\frac{3 \sin \theta - \sin 3\theta}{4} \right) + \frac{1}{3} \left(\frac{3 \sin 3\theta - \sin 9\theta}{4} \right) + \frac{1}{3^2} \left(\frac{3 \sin 9\theta - \sin 27\theta}{4} \right) + \dots + \frac{1}{3^{n-1}} \left(\frac{3 \sin (3^{n-1} \theta) - \sin (3^n \theta)}{4} \right)$$

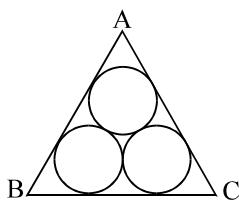
$$= \frac{3 \sin \theta}{4} - \frac{1}{4} \sin 3\theta + \frac{1}{4} \sin 3\theta - \frac{1}{12} \sin (9\theta) + \frac{1}{12} \sin 9\theta - \frac{1}{36} \sin (27\theta) + \dots + \frac{1}{3^{n-2}} \cdot \frac{1}{4} \sin (3^{n-1} \theta) - \frac{1}{3^{n-1} \cdot 4} \sin (3^n \theta)$$

$$= \left[\frac{3 \sin \theta}{4} - \frac{1}{3^{n-1}} \cdot \frac{\sin (3^n \theta)}{4} \right]$$

OTHERS

26. In an equilateral triangle, 3 coins of radii 1 unit each are kept so that they touch each other and also the sides of the triangle. Area of the triangle is :

इकाई त्रिज्या के तीन सिक्के (coins) एक समबाहु त्रिभुज में इस प्रकार से रखे जाते हैं कि वे परस्पर एक दूसरे को तथा त्रिभुज की भुजाओं को स्पर्श करते हैं तो त्रिभुज का क्षेत्रफल होगा :



(A) $4 + 2\sqrt{3}$

(B) $6 + 4\sqrt{3}$

(C) $12 + \frac{7\sqrt{3}}{4}$

(D) $3 + \frac{7\sqrt{3}}{4}$

Ans. B

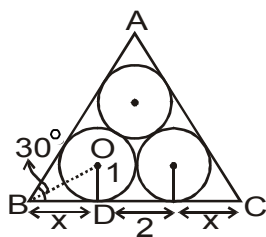
Sol. In $\triangle BOD$

$$\tan 30^\circ = \frac{1}{x}$$

$$\Rightarrow x = \sqrt{3}$$

$$\Rightarrow BC = 2 + 2\sqrt{3}$$

$$\text{area of } \triangle ABC = \frac{\sqrt{3}}{4} (2 + 2\sqrt{3})^2 = 4\sqrt{3} + 6$$



27. Let $\theta \in (0, \pi/4)$ and $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$, then :

माना कि $\theta \in (0, \pi/4)$ तथा $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$ तो :

(A) $t_1 > t_2 > t_3 > t_4$ (B) $t_4 > t_3 > t_1 > t_2$ (C) $t_3 > t_1 > t_2 > t_4$ (D) $t_2 > t_3 > t_1 > t_4$

Ans. B

Sol. $\therefore \theta \in \left(0, \frac{\pi}{4}\right) \Rightarrow \tan \theta < 1$ and $\cot \theta > 1$

let $\tan \theta = 1 - x$ and $\cot \theta = 1 + y$

when $x, y > 0$ and are very small

$$\therefore t_1 = (1 - x)^{1-x}, t_2 = (1 - x)^{1+y}$$

$$t_3 = (1 + y)^{1-x}, t_4 = (1 + y)^{1+y}$$

clearly $t_4 > t_3$ & $t_1 > t_2$

also $t_3 > t_1$

Then $t_4 > t_3 > t_1 > t_2$

28. A circle is inscribed inside a regular pentagon and another circle is circumscribed about this pentagon. Similarly a circle is inscribed in a regular heptagon and another circumscribed about the heptagon. The area of the regions between the two circles in two cases are A_1 and A_2 respectively. If each polygon has a side length of 2 units then which one of the following is true?

एक सम पंचभुज तथा एक सम सप्तभुज के भीतर तथा बाहर वृत्त खींचा जाता है यदि दोनों वृत्त के मध्य का क्षेत्रफल दोनों बहुभुज के लिये यदि A_1 तथा A_2 है तथा प्रत्येक बहुभुज की भुजा 2 सेमी है, तो निम्न में से कौनसा सत्य होगा?

(A) $A_1 = \frac{5}{7} A_2$ (B) $A_1 = \frac{25}{49} A_2$ (C) $A_1 = \frac{49}{25} A_2$ (D) $A_1 = A_2$

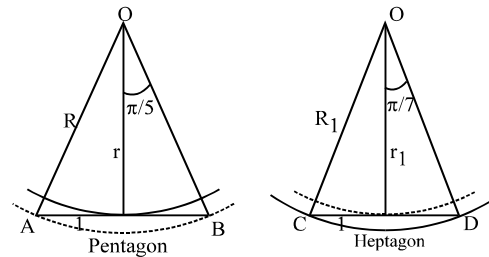
Ans. D

Sol. in 1st case $r = \cot \frac{\pi}{5}$; $R = \operatorname{cosec} \frac{\pi}{5}$

2nd case $r_1 = \cot \frac{\pi}{7}$; $R_1 = \operatorname{cosec} \frac{\pi}{7}$

$$\therefore A_1 = \pi(R^2 - r^2) = \pi \left(\operatorname{cosec}^2 \frac{\pi}{5} - \cot^2 \frac{\pi}{5} \right) = \pi$$

$$\text{|||ly } A_2 = \pi(R_1^2 - r_1^2) = \pi \left(\operatorname{cosec}^2 \frac{\pi}{7} - \cot^2 \frac{\pi}{7} \right) = \pi$$



$\Rightarrow A_1 = A_2$ Ans.

29. If θ and ϕ are acute angles satisfying $\sin \theta = \frac{1}{2}$, $\cos \phi = \frac{1}{3}$, then $\theta + \phi \in$:

यदि $\sin \theta = \frac{1}{2}$ तथा $\cos \phi = \frac{1}{3}$ एवं θ व ϕ न्यूनकोण हो तो $\theta + \phi \in$:

- (A) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$ (B) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ (C) $\left(\frac{2\pi}{3}, \frac{5\pi}{6}\right)$ (D) $\left(\frac{5\pi}{6}, \pi\right)$

Ans. B

Sol. Given

$$\sin \theta = \frac{1}{2}; 0 < \theta < \frac{\pi}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$\text{Given } \cos \phi = \frac{1}{3}; 0 < \phi < \frac{\pi}{2}$$

$$\Rightarrow \phi \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$$

$$\Rightarrow \theta + \phi \in \left(\frac{\pi}{6} + \frac{\pi}{3}, \frac{\pi}{6} + \frac{\pi}{2}\right)$$

$$\Rightarrow \theta + \phi \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$$